
Methods for Interpreting DL models

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1 Introduction

Let's say three people join a data analysis competition. The prize and outcomes based on different participant combinations are shown in the table. Considering all possible combinations, we may wonder about Participant 1's contribution. Assuming equal contributions from all participants, Participant 1's contribution can be determined as follows.

	Value
1st	1000
2nd	700
3rd	500

Table 1: Prize Distribution

Participant	Result
{1,2,3}	1st
{1,2}	1st
{2,3}	3rd
{1,3}	2nd
{1}	3rd
{2}	2nd
{3}	3rd

Table 2: Participant Results

$$\phi_1 = \frac{v(\{1,2,3\})}{3} + \frac{v(\{1,2\})}{2} + \frac{v(\{1,3\})}{2} + v(\{1\})$$
$$\simeq 1533.33$$

Where $v(\cdot)$ is payoff generated by a group of participants in the game.

Let's consider a coalition game with d participants and a value function v (i.e $\langle D, v \rangle : \text{coalition game with } D=[d], v : D \rightarrow \mathbb{R}$). The overall contribution of the i -th participant in the game can be expressed as follows.

$$\phi_i(v) = \sum_{S \subset D, i \notin S} P_S(v(S \cup \{i\}) - v(S)), \quad i = 1, \dots, d$$

where

$$\sum_{S \subset D, i \notin S} P_S = \sum_{s=0}^{d-1} \binom{d-1}{s} P_S = 1, \quad P_S \geq 0, \quad \forall s = 0, \dots, d-1$$

Lloyd Shapley devised a method for fair allocation among all participants in a coalition game $\langle D, v \rangle$ (Shapley et al. (1953)).

$$\phi_i(x) = \sum_{S \subset D \setminus \{i\}} \frac{(d-s-1)!s!}{d!} (v(S \cup \{i\}) - v(S))$$

If the value function is appropriately defined and utilized, it can be applied to various approaches such as regression, classification, and feature selection. In this paper, we will focus solely on regression.

2 Notation and Assumptions

As mentioned earlier, we will address the cases of regression.

$$y = f(x) + \epsilon \quad \epsilon \stackrel{\text{i.i.d.}}{\sim} (0, \sigma^2)$$

The original model f is difficult to handle, so we use an explanation model g which is similar to f but easier to manage. (The following methods all deal with local explanation methods.)

Definition 1. Additive Feature Attribution method

Let $h_x : \{0, 1\}^M \rightarrow \mathbb{X}$, where $h_x(x') = x$

$$g(z') = \gamma_0 + \sum_{i=1}^M \gamma_i z'_i$$

Since handling the original input is challenging, g utilizes simplified input. Therefore, we use map function h_x . If ith value of the input vector is 1, it retrieves the original value, and when it is 0, it retrieves the missing value (or, in some cases, a reference value). For convenience, an example can be drawn from image classification, where g recognizes superpixels as 1. To be a good explanation model, g have to be good approximator for f . (i.e $g(z') \approx f(h_x(z'), z' \approx x')$)

Definition 2. SHAP(SHapley Additive eXplanation values)

Let $\mathbb{X} \subset \mathbb{R}^d$ be feature space, and denote Q as probability measure on $\mathbb{X}$.

$$\begin{aligned} v_S(x) &= \mathbb{E}_Q(f | X_i = x_i, \forall i \in S) \\ &= \mathbb{E}_Q(f(z) | z_s) \\ \phi_i(x) &= \sum_{S \subset D \setminus \{i\}} \frac{(d-s-1)!s!}{d!} (v_{S \cup \{i\}}(x) - v_S(x)) \\ g(z') &= \phi_0 + \sum_{i=1}^d \phi_i z'_i \quad (\phi_0 = \mathbb{E}_Q(f)) \end{aligned}$$

Roughly speaking, SHAP is an Additive Feature Attribution method based on Shapley value coefficients. In the definition above, z_s represents the index set of components that are 1 in the simplified input. Unlike other methods, SHAP possesses desirable properties, which will be discussed in the next chapter.

Similar to the Shapley value, SHAP face the issue of exponentially increasing computation as the number of features grows. Consequently, it becomes challenging to compute the true Shapley value for high-dimensional real data (Castro, J. et al. (2009)). Now, we will introduce some useful definitions for sampling techniques and explanation methods used in deep learning models. The following definitions all assume independence among features.

Definition 3. Sampling Shapley

Let $M = [m]$ be set of all permutations. For each permutation $m \in M$, select the subset $S^m \subseteq D \setminus \{i\}$ with probability $p(s) = \frac{(d-s-1)!s!}{d!}$.

$$\hat{\phi}_i = \frac{1}{M} \sum_{m=1}^M (v_x(S^m \cup \{i\}) - v_x(S^m))$$

Under the assumption of feature independence, the Shapley value can be expressed as the expected value of a specific random variable. Using this property, it can be estimated in the form of Monte Carlo integration, which has desirable properties. Furthermore, Kernel SHAP, which will be introduced next, improves sample efficiency compared to Shapley sampling.

Definition 4. Kernal SHAP

Let $M = [m]$ be set of all permutations. For each permutation m , S^m is sampled from the distribution $p(S)$. Let $\phi = (\phi_1, \dots, \phi_d)^T$, $v_x(z) := v_x(S)$ where z is binary representation of S .

$$p(S) \propto \frac{d-1}{\binom{d}{|S|}(|S|)(d-|S|)}, \quad v_x(\mathbf{1}^d) = f(x), v_x(\mathbf{0}) = E(f)$$

$$\hat{\phi}(x) = \arg \min_{\phi \in \mathbb{R}^d} \frac{1}{M} \sum_{m=1}^M (\phi^T z^m - (v_x(z^m) - v_x(\mathbf{0})))^2 \quad \text{s.t.} \quad \mathbf{1}^T \phi = v_x(\mathbf{1}) - v_x(\mathbf{0}),$$

$$\hat{\phi}(x) = (Z^T Z)^{-1} \left(I - \frac{\mathbf{1} \mathbf{1}^T (Z^T Z)^{-1}}{\mathbf{1}^T (Z^T Z)^{-1} \mathbf{1}} \right) Z^T (V - v_x(\mathbf{0})) + \frac{v_x(\mathbf{1}) - v_x(\mathbf{0})}{\mathbf{1}^T (Z^T Z)^{-1} \mathbf{1}} (Z^T Z)^{-1} \mathbf{1}.$$

Definition 5. LIME(Local Interpretable Model-agnostic Explanation method)

Let G be the set of candidate explanation models. Let $\Omega(g)$ be complexity measure of g , and $L(f, g, \pi_{x'})$ be loss function weighted by local kernel $\pi_{x'}$. LIME minimizes following objective function :

$$\xi = \arg \min_{g \in G} L(f, g, \pi_{x'}) + \Omega(g)$$

LIME is a representative method for local explanations. If the local kernel and loss function in the LIME equation are specifically defined, the solution becomes Kernel SHAP. This will be demonstrated in the next chapter.

Definition 6. DeepLIFT(Deep Learning Important FeaTures)

Let $o = f(x)$ be model output, $\Delta o = f(x) - f(r)$ be difference between output and reference value. Let $\Delta x_i = x_i - r_i$ be difference of input. DeepLIFT attributes to each input x_i a value $C_{\Delta x_i \Delta o}$ satisfying $\sum_{i=1}^d C_{\Delta x_i \Delta o} = \Delta o$.

Deep LIFT is recursive prediction explanation method for deep learning. Deep SHAP is a method that enhances computational efficiency by applying the ideas of DeepLIFT to SHAP. By setting r to $E(f)$ and $C_{\Delta x_i \Delta o}$ to ϕ_i , Deep SHAP is implemented.

3 Key Results

In this section, we provide a useful summary of the concepts defined earlier.

Theorem 1. Fair game

In coalition game $\langle D, v \rangle$, the Shapley value is the unique solution that satisfies followings.

P1 (Efficiency)

$$\sum_{i \in D} \phi_i(v) = v(D)$$

P2 (Symmetry)

If for two players i and j $v(S \cup \{i\}) = v(S \cup \{j\})$ holds for every $S \subseteq D / \{i, j\}$, then $\phi_i(v) = \phi_j(v)$.

P3 (Nullity)

If $v(S \cup \{i\}) = v(S)$ for every $S \subseteq D / \{i\}$, then $\phi_i(v) = 0$.

P4 (Additivity)

For any pair of games $v, w : \phi(v + w) = \phi(v) + \phi(w)$, where $(v + w)(S) = v(S) + w(S)$ for all $S \subseteq D$.

To be considered a fair allocation method (or a fair game), it must satisfy the four properties mentioned above. Lloyd Shapley proved that a solution for a fair game always exists and is unique (Shapley et al. (1953)). Moreover, this solution is the Shapley value. Now, using these properties, we aim to demonstrate the properties of SHAP in the context of additive feature attribution methods.

Theorem 2. Fair game (Additive Feature Attribution method)

Let $f(x)$ original prediction model and $g(x')$ be explanation model by Additive Feature Attribution method.

P1 (Local Accuracy)

$$f(x) = g(x') = \phi_0 + \sum_{i=1}^M \phi_i x'_i$$

The explanation model $g(x')$ matches the original model $f(x)$ when $x = h_x(x')$.

P2 (Missingness)

$$x'_i = 0 \implies \phi_i = 0$$

Missingness constrains features where $x'_i = 0$ to have no attributed impact.

P3 (Consistency)

Let $f_x(z') = f(h_x(z'))$ and $z' \setminus i$ denote setting $z'_i = 0$. For any two models f and f' , if

$$f'_x(z') - f'_x(z' \setminus i) \geq f_x(z') - f_x(z' \setminus i)$$

for all inputs $z' \in \{0, 1\}^d$, then $\phi_i(f', x) \geq \phi_i(f, x)$.

A solution that satisfies the above properties always exists, and it is SHAP (Lundberg, S. et al. (2017)).

4 Proof outlines for the results

As mentioned earlier, both the Shapley value and SHAP encounter computational issues because they require considering the outcomes of all possible combinations ($O(2^d)$). To address this, we aim to present useful theorems for sampling methods designed to mitigate these challenges.

Theorem 3. Asymptotic property of Sampling Shapley

Let ϕ_i be i th true Shapley values, and $\hat{\phi}_i = N^{-1} \sum_{n=1}^N (v_x(S^n \cup \{i\}) - v_x(S^n))$ be i th Sampling Shapley derived by n iterations. Then with regularity assumptions, following holds.

$$\sqrt{N}(\hat{\phi}_i - \phi_i) \xrightarrow{d} N(0, \sigma_i^2)$$

Where σ_i^2 is i th feature variance.

Proof. Let $O : [d] \rightarrow [d]$ be permutation function, and let $\pi(D)$ be the set of all permutation functions. For permutation function O , let $Pre^i(O)$ be the set of predecessors of the permutation O (i.e. $Pre^i(O) = \{O(1), O(2), \dots, O(k-1)\}$ if $i = O(k)$).

$$\begin{aligned} \phi_i(x) &= \sum_{S \subset D \setminus \{i\}} \frac{(d-s-1)!s!}{d!} (v(S \cup \{i\}) - v(S)) \\ &= \sum_{O \in \pi(D)} \frac{1}{d!} (v(Pre^i(O) \cup \{i\}) - v(Pre^i(O))) \\ &= \mathbb{E}(V_{O,x \in \mathbf{X}}) \end{aligned}$$

If we set $V_{O,x \in \mathbf{X}} = v(Pre^i(O) \cup \{i\}) - v(Pre^i(O))$ as random variable with uniform probability, above holds. And Since Sampling Shapley is form of Monte Carlo integration, by Central Limit Theorem, theorem holds. $\square$

Additionally, to improve the sample efficiency of Sampling Shapley, using quasi-random sampling instead of pseudo-random sampling can be one approach. Furthermore, as it is a sampling method, the estimation becomes unstable when the true feature variance (σ_i^2) is high. To address this issue, an alternative is adapting sample (m_i) corresponding to σ_i^2 (Štrumbelj et al. (2014)).

Theorem 4. Deriving Kernal SHAP from LIME

Let g be explanation model from Additive Feature Attribution method. If we set $\pi_{x'}$, L , Ω as followings, then solution for LIME is Kernal SHAP.

$$\begin{aligned}\Omega(g) &= 0 \\ \pi_{x'}(z') &= \frac{(d-1)}{\binom{d}{|z'|}|z'|(d-|z'|)} \\ L(f, g, \pi_{x'}(z')) &= \sum_{z' \in \{0,1\}^d} (f_x(z') - g(z'))^2 \pi_{x'}(z') \quad \text{where } f_x(z') = f(h_x(z'))\end{aligned}$$

Proof. If solution for LIME is true Shapley value, then proof completes.

Let $k(d, 0) = \infty, k(d, d) = \infty, k(d, s) = (d-1)/\binom{d}{s}s(d-s) \quad \forall s = 1, \dots, d-1$.

Since $k(d, 0) = \infty, k(d, d) = \infty$, it gives constraints $f_x(\mathbf{0}) = \phi_0, f_x(\mathbf{1}) = \phi_0 + \sum_{i=1}^d \phi_i$.

Set Y be corresponding vector of $f_x(z')$, $\beta = (\phi_1, \dots, \phi_d)^T$, $X \in \mathbb{R}^{2^d \times d}$ be a corresponding matrix, and $W = \text{diag}(K(d, s_1), \dots, K(d, s_{2^d}))$ be weight matrix, where s_i is sum of all 1 in i th row. Then LIME equation can simply be expressed as :

$$L(f, g, \pi_{x'}(z')) = \arg \min_{\beta \in \mathbb{R}^d} (Y - X\beta)^T W (Y - X\beta) \quad \text{s.t } \mathbf{1}^T \beta = f(x) - \mathbb{E}(f(X))$$

Let $\hat{\beta} = (X^T W X)^{-1} (X^T W Y)$. If $\hat{\beta}$ satisfies $\mathbf{1}^T \beta = f(x) - \mathbb{E}(f(X))$ then it is solution. Since $k(d, 0) = \infty, k(d, d) = \infty$, $(X^T W X) = \frac{1}{d-1} I + cJ$ as $c \rightarrow \infty$, and $(X^T W X)^{-1} = I + \frac{1}{d-1} (I - J)$ as $c \rightarrow \infty$.

$$\begin{aligned}\hat{\beta}_j &= e_j^T \hat{\beta} \\ &= e_j^T (X^T W X)^{-1} (X^T W Y) \\ &= (e_j^T - \frac{1}{d-1} (\mathbf{1}^T - e_j^T)) X^T W Y \\ (e_j^T - \frac{1}{d-1} (\mathbf{1}^T - e_j^T)) X^T W &= (k(d, s_i) (\mathbb{I}(X_{ij} = 1) - \frac{1}{d-1} (s_i - \mathbb{I}(X_{ij} = 1)))_{i=1, \dots, 2^d} \\ \hat{\beta}_j &= \sum_{S \subset D \setminus \{i\}} \frac{(d-s-1)! s!}{d!} (v(S \cup \{i\}) - v(S))\end{aligned}$$

□

The above derivation is significant in that it explains how Kernel SHAP can be derived from the LIME equation. However, there is a key difference between LIME and Kernel SHAP. While LIME assigns larger weights to samples closer to the original input, Kernel SHAP places higher weights on samples where $|z'|$ takes very small or very large values. Roughly speaking, this difference reflects the idea that the information content is higher when $|z'|$ is either very small or very large (Lundberg, S. et al. (2017)).

5 Application

In this section, we aim to examine how Shapley values function as an explanation model and how they operate within a DNN model. Additionally, we will discuss the distinguishing features of DNN compared to other models and address the challenges that arise when dealing with complex models like DNN. The data used in this analysis was sourced from the UCI Machine Learning Repository (Dataset A) and analyzed using Python. The primary library utilized for calculating Shapley values was the SHAP library.

5.1 How it work as Interpretation Model

Model: DNN Prediction: $p(\text{survived} = \text{yes} \mid x) = 0.503$ Actual Class: yes			Model: NB Prediction: $p(\text{survived} = \text{yes} \mid x) = 0.671$ Actual Class: yes		
Feature	Value	Contribution	Feature	Value	Contribution
Class	3rd	-0.289	Class	3rd	-0.078
Age	adult	-0.042	Age	adult	-0.066
Sex	female	0.162	Sex	female	0.23

The table above presents the analysis results using the well-known Titanic dataset. A fully connected DNN with 3 hidden layers and 12 nodes per layer was fitted, along with a Naive Bayes model. For each fitted model, the Shapley value was calculated for a randomly selected test data point, and the results are shown in the table.

First, both models predicted that the passenger would survive, which was correct. However, the Naive Bayes model expressed greater confidence in the passenger's survival compared to the DNN model. For the DNN model, it was determined that the environment of being a third-class passenger contributed significantly to reducing the survival probability. On the other hand, the passenger's gender (female) was interpreted as a factor that contributed positively to the survival probability.

For the Naive Bayes model, while the direction of the feature contributions was consistent with the DNN model, the magnitude of the contributions differed. This demonstrates how Shapley values can show the extent to which each feature contributes to the prediction value. Such interpretability makes Shapley values crucial in fields requiring explanation, such as medicine, economics, and criminology. Additionally, it is significant in that it can address the chronic issue of complex models being "black boxes."

5.2 Computational issue

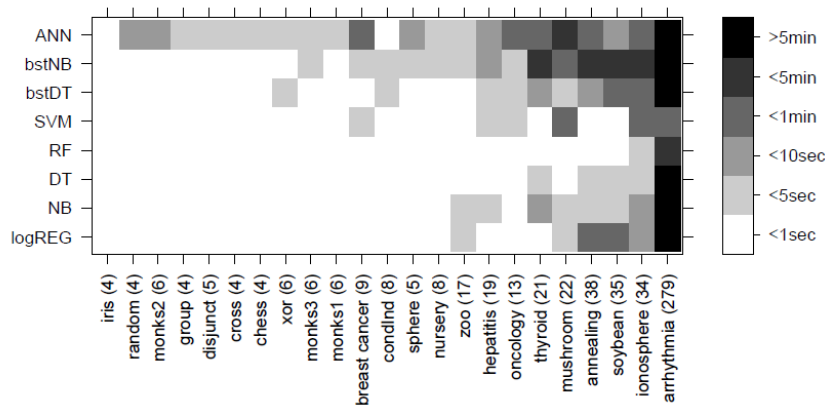


Figure 1: Running times across all data set and models.

The figure above illustrates the time required to compute Shapley values for each model and dataset obtained from source (Štrumbelj et al. (2014)). As shown in the figure, the running time increases significantly with a higher number of features or when using a complex model. Therefore, calculating true Shapley values for real-world data with many features becomes extremely challenging. Furthermore, the use of complex models exacerbates this computational issue. Thus, for deriving interpretation to real data based prediction, Shapley sampling method is needed.

5.3 The challenges of complex models and methods for their detection

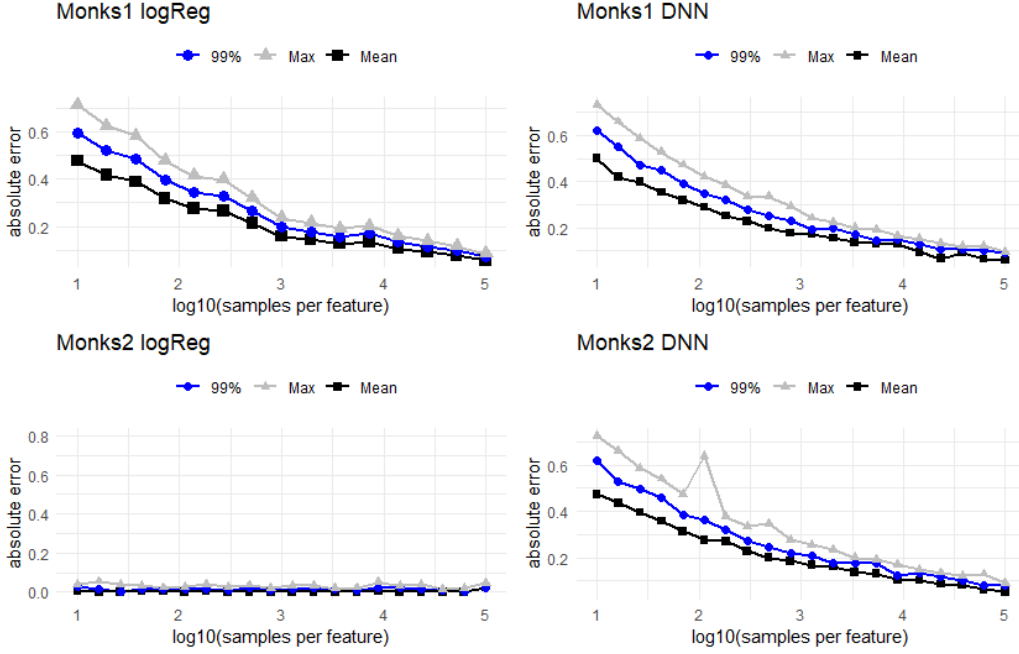


Figure 2: Absolute difference between approximated and actual contribution

This subsection aims to verify whether the Shapley sampling method converges to the true Shapley values and to address the challenges of complex models along with methods for detecting these issues. The analysis uses the Monks1 and Monks2 datasets from the UCI Machine Learning Repository.

To briefly describe the datasets: Monks1 contains a data pattern where the target value is 1 if either ($\text{attr1} = \text{attr2}$) or ($\text{attr5} = 1$). In contrast, Monks2 consists of random noise with no inherent data knowledge. As shown in the table above, the results demonstrate that as the number of samples increases, the approximation becomes closer to the true values.

According to the results in (Štrumbelj et al. (2014)), the DNN effectively captures these two specific characteristics of the Monks1 dataset, functioning as a good predictor. When compared to the Naive Bayes model, it is evident that Shapley values do not merely reveal the underlying characteristics of the data but also provide insights into what the model learns from the data.

Furthermore, as observed in the table, the logReg model applied to Monks2 data, which lacks any inherent knowledge, exhibits an absolute error close to zero. This outcome reflects that simple models extract little to no knowledge from such data. On the other hand, as shown in the Monks2 DNN results, the DNN model arbitrarily identifies patterns and learns from data with no meaningful knowledge, highlighting the issue of overfitting in complex models.

To detect this issue, a useful approach is to incorporate a simple model and compare the trends in absolute error, as demonstrated above. This comparative analysis can serve as an effective method for identifying overfitting in complex models.

6 Conclusion

Despite its high accuracy, the Shapley value can serve as an effective interpretability metric for deep learning models, which often carry the risks of being "black box." However, previous discussions have assumed feature independence, meaning that if this assumption is significantly violated, issues may arise (Control SHAP).

Additionally, alternative sampling methods have been introduced to address computational challenges. As a result, research is being conducted to better control sample variation in Shapley value computations, such as Rank SHAP and Max SHAP.

In the past, Shapley values only explained the extent to which a variable influenced the prediction value, making it difficult to discuss variable importance rankings or feature elimination methods. However, by carefully specifying the value function so that the Shapley value is e_value (Vovk, V. et al. (2021)) it becomes possible to use the Shapley value as a criterion for variable selection as well.

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